A Flavonoid-inspired molecular generation

Corpus preparation
ChEMBL · PubChem · seeds
1,689 curated molecules
Bemis–Murcko scaffold split

Train	Validation	Test
1,353	188	148

Transformer training
Decoder-only SMILES model
8,118 training sequences
Training augmentation only
Validation PPL 1.7366
Test PPL 1.5854

De novo generation
30,000 sampled SMILES

Valid	**21,715**
Unique	**12,436**
Novel	**12,136**

Medicinal chemistry filtering
Physicochemical limits
QED ≥0.50 · SA ≤6.0
Alerts and flavonoid domain
2,371 candidates

B Structure-based prioritization

Haemoglobin model
PDB 6XD9
Protein preparation + grid
VZHE-039 redocking
RMSD 0.7754 Å

Hierarchical Glide screening

XP	SP	HTVS
955	1,329	3,337

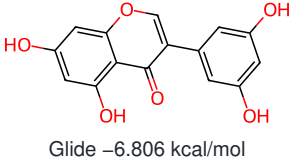
Conformations entering each stage

Ligand preparation
2,371 candidates
3,337 conformations
LigPrep

Five candidates from XP screening
Prime MM-GBSA · PLIP · ADMETlab 3.0
Voxelotor as contextual reference

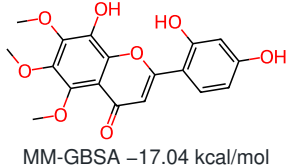
C Molecular dynamics follow-up

FGHBS_00418



Glide –6.806 kcal/mol

FGHBS_01243



MM-GBSA –17.04 kcal/mol

Molecular dynamics
Desmond · 200 ns per complex
Both candidates + voxelotor reference
One trajectory for each complex
RMSD · RMSF · ligand properties
Protein–ligand contacts